Real vs. Fake Face Detection — Baseline Results Summary

Master 2 Thesis — First Working Notebook (Deepfake Detection Track)

1. Objective

Build and validate a first end-to-end pipeline for digital face forgery (deepfake) detection: load real vs. AI-generated face images, fine-tune a pretrained CNN to classify them, and evaluate on a held-out test set — before iterating toward the state-of-the-art methods identified in the literature review.

2. Method

- Model: ResNet18 pretrained on ImageNet; backbone frozen, only the final classification layer fine-tuned (linear probe).
- Dataset: "140k Real and Fake Faces" (Kaggle) — real photographs (FFHQ) vs. StyleGAN-generated faces. 100,000 train / 20,000 validation images, perfectly class-balanced.
- Training: 5 epochs, batch size 64, Adam optimizer (lr=1e-3), on a Colab T4 GPU (~31 min total).
- Evaluation: held-out test split of 20,000 images (10,000 real / 10,000 fake), never seen during training.

3. Results

Metric	Value	Metric	Value
Test Accuracy	80.8%	Precision (both classes)	0.81
Test AUC	0.888	Recall (both classes)	0.81
Best Val. Accuracy	81.1%	F1-score (both classes)	0.81

Confusion matrix (test set, n = 20,000):

	Predicted: Fake	Predicted: Real
True: Fake	8050	1950
True: Real	1896	8104

4. Interpretation

- The model correctly separates real from GAN-generated faces well above chance, with balanced performance across both classes (no bias toward predicting one class over the other).
- AUC (0.888) noticeably exceeds accuracy (80.8%), indicating good underlying separability; the fixed 0.5 decision threshold is not yet optimal.
- Accuracy plateaued after epoch 1 (~80%), consistent with only the final layer being trainable — the frozen backbone limits how much the model can specialize to forgery-specific artifacts.

5. Limitations & Next Steps

- Trained/tested on a single generator family (StyleGAN); generalization to other GANs, diffusion models, or video-based face-swap deepfakes (e.g., FaceForensics++) is untested.
- Next: fine-tune deeper layers (unfreeze last residual block) to push past the current plateau.
- Next: cross-dataset evaluation on a different fake-face source to obtain the thesis's first generalization result (Section 8 of the SOTA review).

Code & results (Colab notebook, outputs included): [GitHub repository link]